



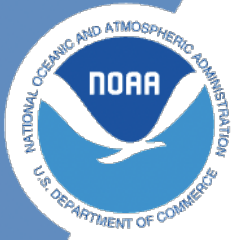
NOAA Technical Memorandum NMFS-XXX-##

NOAA quarto book

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National Oceanic and Atmospheric
Administration
National Marine Fisheries Service
Northwest Fisheries Science Center



**NOAA
FISHERIES**

NOAA quarto book

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1 Introduction

Here we will be displaying various data summaries and analyses related to dolphin management in the U.S. South Atlantic.

1.1 Load Libraries

The documentation of this data exploration was developed in R, using R Studio and Quarto. The following R libraries were also used. The `pals` package allows for improved visualization with different color palettes (**pals?**). The `rerddap` package allows for efficient downloads of physical data from the ERDDAP server (**rerddap?**).

2 Chapter 1 - NOAA Fisheries landings data

Here we will explore commercial and recreational landings data accessed through the FOSS (Fisheries One Stop Shop) data portal. The advantage of this portal is that commercial and recreational data are available side-by-side in the same units (pounds landed).

2.1 Data access and upload

Data are downloaded from the NMFS FOSS site: [<https://www.fisheries.noaa.gov/foss/f?p=21>]. For the Data Set, we choose both commercial and recreational. For the Years, we select all years from 1980 to 2021. For NMFS Region, we select the "NMFS Regions" button and then select the Gulf, South Atlantic, Middle Atlantic and New England from the box. For the Species, we select "Dolphinfish" and "Dolphinfish**." For Report Format, we choose TOTALS BY YEAR/REGION. Hitting the "RUN REPORT" button then generates the data needed for the subsequent analysis, which can be downloaded under the "Actions" button.

We then read in the data file and clean it for analysis (removing the commas from the numbers so that R recognizes the values as numbers). We output the file header to make sure it was read in and that the numbers were converted correctly.

	Year	Region.Name	Pounds	Dollars	Collection	Metric.Tons
1	2000	Gulf	582,576	689,637	Commercial	264
2	2000	Gulf	5,641,491		Recreational	2,559
3	2000	New England	6,545	12,414	Commercial	3
4	2000	South Atlantic	537,077	821,743	Commercial	244
5	2000	South Atlantic	16,049,049		Recreational	7,280
6	2000	Middle Atlantic	36,048	65,004	Commercial	16

	Year	Region.Name	Pounds	Dollars	Collection	Metric.Tons
1	2000	Gulf	582576	689637	Commercial	264
2	2000	Gulf	5641491	NA	Recreational	2559

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3	2000	New England	6545	12414	Commercial	3
4	2000	South Atlantic	537077	821743	Commercial	244
5	2000	South Atlantic	16049049	NA	Recreational	7280
6	2000	Middle Atlantic	36048	65004	Commercial	16

2.2 Summarize the landings

We then summarize the landings (in pounds) by year, sector, and region. The result is a new data frame with each row representing a year and each column representing the total landings for each region and sector.

, , Gulf

	Recreational	Commercial
1980	0	94984
1981	1559544	73075
1982	4197503	162457
1983	2162936	190808
1984	1189573	282976
1985	2542089	269477

, , Middle Atlantic

	Recreational	Commercial
1980	0	400
1981	0	0
1982	2097	200
1983	45927	1400
1984	0	1700
1985	113666	5000

, , New England

	Recreational	Commercial
1980	0	200
1981	0	0
1982	0	0
1983	0	400
1984	0	400

2 Chapter 1 - NOAA Fisheries landings data

1985 0 4800

, , South Atlantic

	Recreational	Commercial
1980	0	76042
1981	8421898	59037
1982	7573035	143960
1983	9134022	127986
1984	7265430	158901
1985	6684821	142684

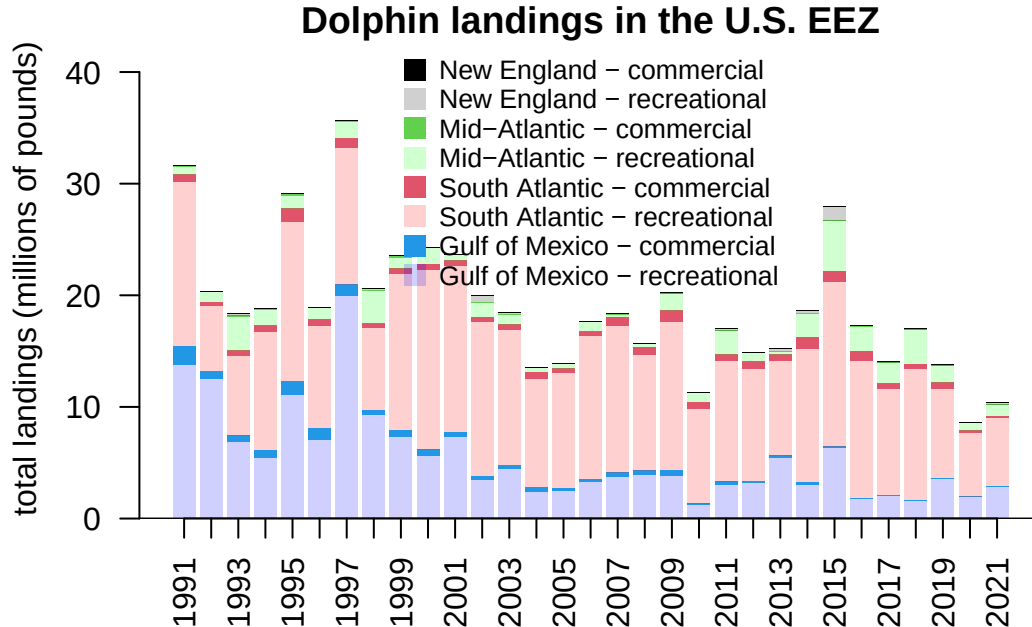
	rec_Gulf	com_Gulf	rec_SA	com_SA	rec_MA	com_MA	rec_NE	com_NE
1980	0	94984	0	76042	0	400	0	200
1981	1559544	73075	8421898	59037	0	0	0	0
1982	4197503	162457	7573035	143960	2097	200	0	0
1983	2162936	190808	9134022	127986	45927	1400	0	400
1984	1189573	282976	7265430	158901	0	1700	0	400
1985	2542089	269477	6684821	142684	113666	5000	0	4800
1986	5540003	492212	6190187	189747	125653	4200	0	200
1987	4209638	403703	6883546	212905	121433	13400	0	1100
1988	1014312	491610	3134070	212709	130240	26600	2196	17800
1989	12551461	1029628	13273504	426392	1386752	81700	437	15300
1990	13389540	1098819	9283973	514980	624928	69106	2714	14233
1991	13760600	1674124	14756442	650333	722090	90722	6294	9816
1992	12481422	714369	5893700	326831	860112	74423	1411	10315
1993	6873820	596967	7077105	520512	3001335	99420	200927	27069
1994	5448709	679903	10554836	618614	1400635	101044	12586	12328
1995	11142952	1177325	14318412	1164608	1096473	198425	24061	13811
1996	7086006	1047020	9128895	548682	1077364	43581	0	7788
1997	19958088	989648	12284366	877303	1422057	105559	0	13265
1998	9324186	422501	7312090	406368	2920384	86502	120322	11553
1999	7329953	612445	13973364	504263	1005852	98927	4458	5990
2000	5641491	582576	16049049	537077	1433991	36048	0	6545
2001	7355385	368503	14945284	470897	477081	51342	5189	29134
2002	3513226	290601	13794853	436472	1295470	83534	600948	14836
2003	4481937	310836	12112718	459351	929763	46851	129513	21275
2004	2402315	435341	9707678	555973	427145	31916	0	29042
2005	2537132	207695	10299670	376989	437472	19703	0	19982
2006	3290474	224668	12805952	433121	887684	27724	0	19645
2007	3754513	370423	13164377	739862	241212	77553	19542	40141

2 Chapter 1 - NOAA Fisheries landings data

2008	3937153	385011	10417536	627506	264017	53013	0	13448
2009	3860527	497246	13251102	1006948	1538112	49946	0	76449
2010	1216198	112802	8543542	571983	813738	34578	0	12910
2011	3023914	312567	10774812	544333	2173136	14322	187289	22816
2012	3229090	148474	9983276	704483	755558	42421	26892	43955
2013	5425636	191201	8483569	584102	302421	39214	226988	16657
2014	3074373	207261	11925345	1042822	2129222	48462	183673	51810
2015	6286577	155696	14821772	920772	4501793	34858	1198628	31995
2016	1777348	37733	12357817	828435	2212513	61336	16499	15747
2017	2064106	68260	9457460	533082	1820541	16179	48607	30969
2018	1626095	57052	11713866	435290	3104018	20514	97107	12667
2019	3567170	85077	7967978	563708	1518629	47981	66	16918
2020	1952241	26653	5653294	260503	679556	14040	0	1880
2021	2893817	27212	6061024	177263	1051785	25799	123038	11421

2.3 Plot the summary

Now we plot the data, using similar color schemes by region. Because the recreational data have high imprecision prior to the 1990s we plot from 1990 to the last year of data.

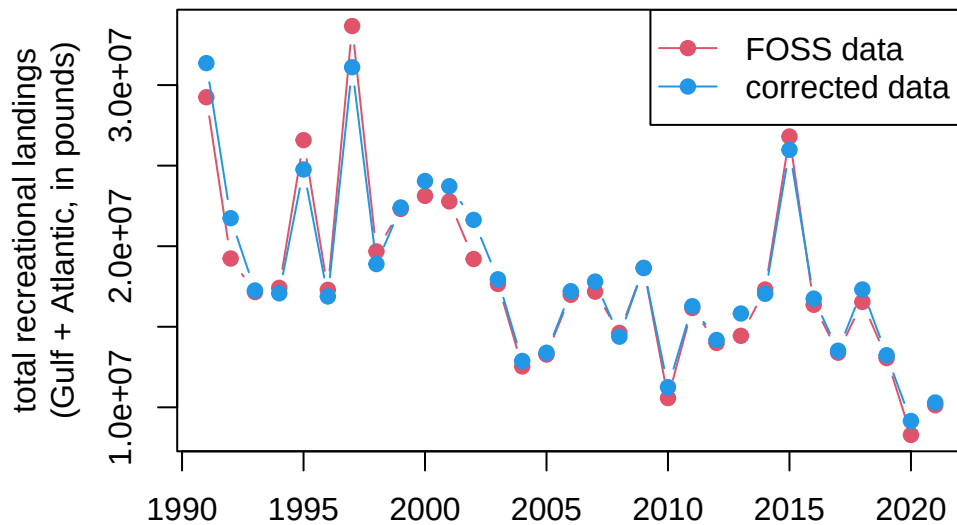


3 Updated Gulf/Atlantic parsing of data

There are some issues with the way that the FOSS system splits out Gulf and Atlantic catches by Monroe County, which is on the border between the two jurisdictions. We received data from Mike Larkin of SERO which has the correct split out of data for Monroe County between the Gulf and Atlantic. We compare the total recreational catch for the Gulf and Atlantic between the FOSS data and the updated data. The numbers are very close; the average difference from the FOSS data is 2%.

Mode	TRUE
logical	31

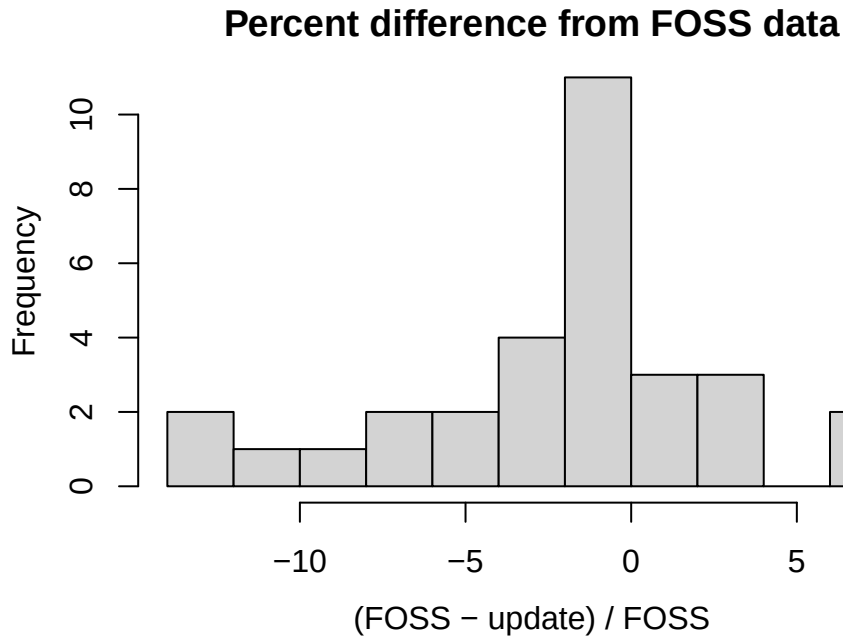
Comparison of FOSS data with corrected split by Monroe



[1] -2.023886

3 Updated Gulf/Atlantic parsing of data

[1] -13.008552 7.587789



	rec_Gulf	com_Gulf	rec_SA	com_SA	rec_MA	com_MA	rec_NE	com_NE
2016	1777348	37733	12357817	828435	2212513	61336	16499	15747
2017	2064106	68260	9457460	533082	1820541	16179	48607	30969
2018	1626095	57052	11713866	435290	3104018	20514	97107	12667
2019	3567170	85077	7967978	563708	1518629	47981	66	16918
2020	1952241	26653	5653294	260503	679556	14040	0	1880
2021	2893817	27212	6061024	177263	1051785	25799	123038	11421

	rec_Gulf	com_Gulf	rec_SA	com_SA	rec_MA	com_MA	rec_NE	com_NE
2016	751499	37733	13766851	828435	2212513	61336	16499	15747
2017	868489	68260	10777104	533082	1820541	16179	48607	30969
2018	508570	57052	13603875	435290	3104018	20514	97107	12667
2019	1304865	85077	10410599	563708	1518629	47981	66	16918
2020	555695	26653	7916891	260503	679556	14040	0	1880
2021	1174412	27212	7950716	177263	1051785	25799	123038	11421

We replace the FOSS values for Gulf and South Atlantic recreational catches with these new values and replot the full data set.

3 Updated Gulf/Atlantic parsing of data

Dolphin landings in the U.S. EEZ -- correct Monroe Count

